

AREA III:

AB Structures and Environment Engineering and Bioprocess Engineering and Allied Subjects

TOPIC 3:

Bioprocess and Food Engineering

3.2 Apply the theories and principles of psychrometrics, heat transfer, thermodynamics and others

Number of items: 5-6 questions

Coverage: Psychrometry, Heat Transfer, and Thermodynamics Computation

Psychrometry

This deals with the thermodynamic properties of mixed air and water vapor, often used in HVAC (heating, ventilation, and air conditioning) systems to control environmental conditions.

Sample questions

$$J = Nm$$

$$Pa = N/m^2$$

What is the density of dry air at 93 kPa and 45°F?

convert to K

a. 1.28

b. 1.16

c. 1.51

d. 1.39

$$PV = mRT$$

$$\frac{m}{V} = \frac{P}{RT}$$

$$= \frac{(93000 \frac{N}{m^2})}{(287 \frac{Nm}{kgK}) (280.22K)} \left[\frac{kg}{m^3} \right]$$

$$\rho = \frac{m}{V} = 1.16 \frac{kg}{m^3}$$

Sample questions

Consider 250 m³ of moist air at 1 bar, 35°C, 80% RH. If saturation pressure of water is 5.6 bar at 35°C, then partial pressure of dry air is:

- a. 5.48 bar
- b. 448 kPa
- c. 548 kPa
- d. 8.44 bar

$$\begin{aligned} RH &= \left(\frac{P_p}{P_s} \right) (100) \\ 80 &= \left[\frac{P_p}{5.6} \right] (100) \end{aligned} \quad \begin{aligned} &\rightarrow P_p = 4.48 \text{ bar} \\ &\quad \left(4.48 \text{ bar} \right) \left(\frac{100 \text{ kPa}}{1 \text{ bar}} \right) \\ &= 448 \text{ kPa} \end{aligned}$$

Sample questions

A room is filled with saturated moist air at 29°C and a total pressure of 102 kPa and actual partial pressure of 0.4 bar. If the mass of dry air in the room is 150 kg, the mass of water vapor is:

$$HR = \frac{m_v}{m_a} = 0.622 \left[\frac{P_p}{P_t - P_p} \right] \Rightarrow \frac{m_v}{150 \text{ kg}} = 0.622 \left[\frac{40 \text{ kPa}}{102 - 40} \right]$$
$$m_v = 60 \text{ kg}$$

Sample questions

A room contains 55 kg of dry air and 0.8 kg of water vapor. The total temperature and pressure in the room are 32°C and 1 bar. Given that the saturation pressure for water at 32°C is 0.5 bar, the relative humidity of air in the room is:

$$\text{RH} = 4.56\%$$

① $\frac{m_v}{m_a} = 0.622 \left[\frac{P_p}{P_t - P_p} \right] = \frac{0.8}{55} = 0.622 \left[\frac{P_p}{100 \text{ kPa} - P_p} \right]$

② $\text{RH} = \left(\frac{P_p}{P_s} \right) (100) = \frac{2.284 \text{ kPa}}{(0.5 \times 100)} \times 100$

Sample questions

A room contains air at 35°C and a partial pressure of 3.7 kPa with a relative humidity of 78 percent. Given that the humidity ratio is 0.015, the total pressure is:

$$0.015 = 0.6222 \left[\frac{3.7 \text{ kPa}}{P_T - 3.7} \right]$$

$$P_T = 157.18 \text{ kPa}$$

Sample questions

On a psychrometric chart, cooling and humidification process appears as a line that is:

- a. diagonal upwards to the right — heating & humidification
- b. diagonal downwards to the left — cooling & dehumidification
- c. diagonal upwards to the left
- d. diagonal downwards to the right — heating & dehumidification

Sample questions

On a psychrometric chart, heating and dehumidification process appears as a line that is:

- a. diagonal upwards to the right
- b. diagonal downwards to the left
- c. diagonal upwards to the left
- d. diagonal downwards to the right

Sample questions

Condensation occurs at:

- a. Saturation Point
- b. Vaporization Point
- c. Dew Point
- d. Fusion Point

Sample questions

This states that the mixture pressure is equal to the sum of the partial pressure to the sum of the partial pressure of the constituents.

- a. Dalton's Law
- b. Weibach's Law
- c. Newton's Law
- d. Brandtt's Law

THANK YOU!!!

Proverbs 3:5-6

**“Trust in the Lord with all your heart
and lean not on your own
understanding; in all your ways
submit to Him, and He will make
your paths straight”**